

AN OVERVIEW OF THE ARCHAEOLOGY OF SAN NICOLAS ISLAND, SOUTHERN CALIFORNIA

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ABSTRACT

San Nicolas Island, the outermost of the southern Channel Islands, has been investigated for over 100 years, but little of it has been adequately analyzed and published. The available reports, which are mostly unpublished, show that the island was occupied as early as 6,000 years ago. Midden samples reveal a wide variety of shellfish, fish, and sea mammal bone indicating a marine subsistence focus. Burials and a variety of artifacts, such as pestles, bowl fragments, bone implements, basketry impressions, cobble tools, shell beads, and fishhooks, reflect a broad range of cultural activities and suggest permanent settlement. These data form the basis for planning further work in terms of contemporary research goals to better understand the island's prehistory and human adaptations to this restricted environment.

INTRODUCTION AND BACKGROUND

San Nicolas Island is located approximately 120 km southwest of Los Angeles and 102 km from the nearest point of land at Point Mugu, California (Fig. 1). It is 14 km long and 5 km wide. At the time of European contact, the island was occupied by Indians who probably spoke one of the Cupan languages in the Takic family, which is part of the Uto-Aztecan linguistic stock. This assumption is based on a few words that were recorded as spoken by the last survivor, a woman who was left behind on San Nicolas Island when the remaining small population was removed to the mainland in 1835 (Bean and Smith 1978; Eisen 1905; Meighan and Eberhart 1953).

There was archaeological interest in San Nicolas Island as early as the 1870s. Investigators such as Paul Schumacher, Léon de Cessac, Stephen Bowers, Bruce Bryan, Malcolm Rogers, Arthur Woodward (n.d.), and Phil Orr (1945 and 1946) have all worked on the island. Numerous surveys and excavations have been conducted, though for the most part these investigations have gone unreported. As Meighan and Eberhart (1953:113) remarked,

None of these collections have been published except for brief notes on exceptional pieces. Although many of the collections were made by untrained individuals in the days before scientific records were kept, it is nonetheless regrettable that despite some 15,000 specimens from San Nicolas preserved in American museums so little is known of the island's archaeology.

Meighan and Eberhart's paper "Archaeological Resources of San Nicolas Island, California" (1953) marked a major shift in archaeological studies on San Nicolas Island. As the title implies, the shift was toward looking at archaeological sites as "resources" which could be

used to answer specific research questions. The majority of subsequent work has been question-oriented and the level of documentation has increased dramatically.

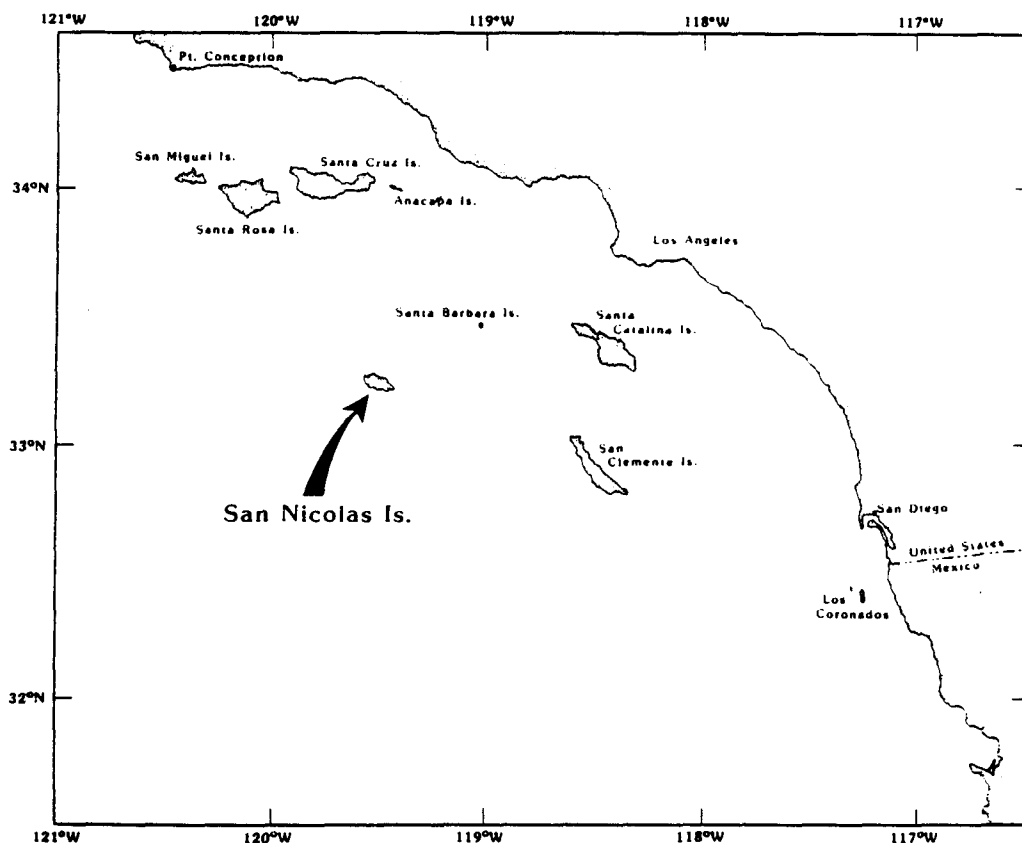


Fig.1. Location of San Nicolas Island.

Fred Reinman began work at San Nicolas Island during the late 1950s. His investigations include the excavation of six burial sites (Reinman and Townsend 1960), a survey of the upper plateau of the island (Reinman 1962), and a series of test excavations to clarify cultural sequences at SNI-11 and SNI-16 (Lauter n.d.a and n.d.b; Reinman n.d.a and n.d.b; Reinman and Lauter n.d.a). In addition, an intensive, systematic survey of the entire island was conducted by Reinman in 1983-1984 as a resource management project. The purpose of this survey was to locate all of the visible archaeological resources, provide information regarding integrity and eligibility for the National Register of Historic Places, and provide recommendations for further research and management (Reinman and Lauter n.d.b).

In spite of all of the attention that the island's archaeological resources have received, only ten of the 500 known sites have been excavated and reported on specifically. Of the ten, one is located on the central plateau, two are on the southern shore, and seven are on the northwest shore (Fig. 2).

Although ten sites have been excavated, five of these investigations were limited to removal of exposed burials; the midden areas were not tested (Reinman and Townsend 1960). Sites SNI-38 and SNI-51 were tested by Rozaire (1959) for the express purpose of

locating woven materials for an analysis of the weaving technique; some midden analysis was performed incidentally. Three sites, SNI-11, SNI-16, and SNI-79, have been examined primarily for midden constituents (Reinman 1964). Of these, the most fully reported to date is SNI-11, which was the subject of two Master's theses (Clevenger n.d.; Bleitz-Sanburg n.d.a), part of a dissertation (Salls n.d.b), and a site report (Reinman n.d.a). SNI-16 was the basis for one Master's thesis (Lauter n.d.a) emphasizing chronology.

This lack of reporting has resulted in a very limited knowledge of the island's archaeology. Future research should increase understanding of the cultural processes. To this end, a preliminary research design has been developed, and archaeological investigations are currently being conducted by California State University, Los Angeles (CSULA), under the direction of Patricia Martz. Excavations were also conducted by California State University, Fullerton, and Chaffey College in 1991, and these data will be analyzed at CSULA and integrated into a comprehensive research design. This review of the past archaeological work on San Nicolas Island reflects the current interest in developing new research into the prehistory of the island and provides the background for these studies.

SNI-11, Thousand Springs Site

The Thousand Springs site is the only extensively excavated site on the island that has been adequately reported. The site is located on the island's northwest coast near one of the main water sources, near a rocky intertidal zone. SNI-11 is a large dune site measuring some 450 by 200 m. The site is subdivided into four mounds designated A, B, C, and D. At least three strata are observable in Mound B. The site was tested on two occasions – first by the UCLA Archaeological Survey in 1959 (Reinman and Townsend 1960) and then by CSULA in 1977-79 (Reinman n.d.a). The site is still considered to have excellent research potential.

A series of column samples were taken in 1959 by the UCLA Archaeological Survey. The depth of the midden varied in the units, ranging from 38 to 56 cm below the present surface. Three samples from three locations along the western edge of Mound A were examined for midden constituents. Analysis revealed that four main shellfish species were present in the midden: abalone, tegula, mussel, and sea urchin. The exploitation of these species changed through time. The procurement of mussels remained relatively constant, while abalone usage increased and tegula usage decreased. The usage of sea urchin increased and then decreased through time and seemed to fluctuate more than the others (Reinman 1964).

In 1977-1979 CSULA excavated fifteen 1.5- by 1.5-m test units. The material was screened through 1/4-inch mesh. A number of radiocarbon dates have been processed from this site making it the best-dated site on the island (Table 1).

Based on a preliminary analysis prior to the submission of the radiocarbon samples, Reinman suggested that the artifacts indicate a good stratigraphic context with Late Period fish hooks in the upper Stratum I, and bone fish gorges, leaf-shaped projectile points, and bifaces in the lower strata (n.d.a). However, a review of the radiocarbon dates reveals some stratigraphic inconsistencies and indicates the need for a re-evaluation of the sampling and processing procedures, as well as a better understanding of the post-depositional processes of a sand dune midden site. For example, Mound B, Stratum I, has a 2469-2046 B.C. (4419-3996 B.P.) date and an A.D. 1280-1387 (670-536 B.P.) date at 10-20 cm. A date of 4894-4846 B.C. (6844-6796 B.P.) overlies that of 2869-2701 B.C. (4819-4651 B.P.) in Stratum III. Mound A contains only Stratum I, and the base of the mound, dated at 399-123 B.C. (2349-2073 B.P.), is overlain by a date of 3036-2896 B.C. (4986-4846 B.P.) at the 30-40 cm level.

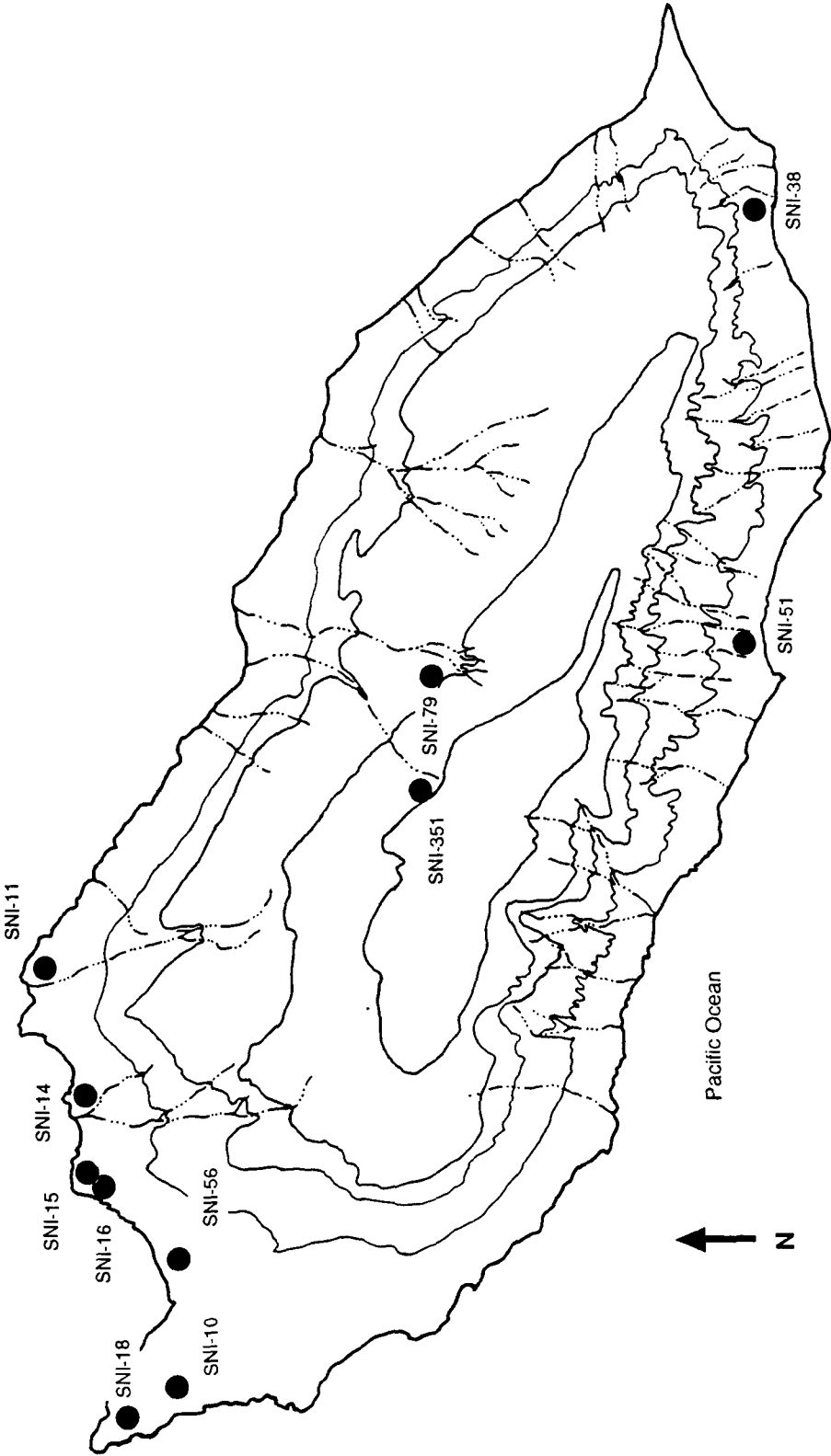


Fig. 2. San Nicolas Island archaeological sites.

Table 1. Radiocarbon Dates from San Nicolas Island

Site	Sample	Location	Unit	Depth	Material	C-14 years	Calibration date BP*
SNI-11	UCLA-2559A	Mound A	3S/1.5E	30-40cm (Stratum I)	Charcoal	3725 ±80	4230 (4088) 3982
SNI-11	UCLA-2559B	Mound A	3S/1.5E	70-80cm (Stratum I)	Charcoal	2460 ±100	2739 (2704 - 2486) 2349
SNI-11	UCLA-2559C	Mound A	3S/1.5E	130-140cm (Stratum I)	Charcoal	4330 ±80	4986 (4870) 4846
SNI-11	GaK-8204	Mound A	3S/1.5E	150-170cm (Stratum I)	Charcoal	2220 ±110	2349 (2311 - 2207) 2073
SNI-11	IVC-82	Mound B	4.5S/22.5E	10-20cm (Stratum I)	Charcoal	960 ±46	934 (920) 795
SNI-11	GaK-8205	Mound B	1.5S/3E	10-20cm (Stratum I)	Charcoal	3820 ±120	4419 (4237) 3996
SNI-11	IVC-81	Mound B	1.5S/3E	20-30cm (Stratum I)	Charcoal	650 ±45	670 (660) 563
SNI-11	IVC-44	Mound C	3N/64.5E	60-67cm (Stratum I)	Charcoal	1559 ±120	1570 (1485 - 1420) 1330
SNI-11	IVC-45	Mound D	1.5N/39E	20-30cm (Stratum I)	Charcoal	573 ±109	670 (553) 510
SNI-11	GaK-8206	Mound B	0N/31.5W	10-20cm (Stratum II)	Charcoal	3430 ±130	3859 (3692) 3492
SNI-11	UCLA-2559E	Mound B	7.5S/3E	60-70cm (Stratum II)	Charcoal	3930 ±125	4539 (4413) 4229
SNI-11	UCLA-2559D	Mound B	0N/31.5W	140-150cm (Stratum III)	Charcoal	5955 ±120	6983 (6844 - 6796) 6679
SNI-11	GaK-8207	Mound B	0N/31.5W	170-190cm (Stratum III)	Charcoal	4160 ±140	4869 (4819 - 4651) 4455
SNI-16	UCLA-165	Burial area	?	?	Haliotis	3300 ±100	2959 (2832) 2739
SNI-16	IVC-43	Hearth	3.5N/27E	60-70cm	Charcoal	3682 ±148	4249 (4074 - 3991) 3839
SNI-18	UCLA-164	Burial area	?	?	Haliotis fishhook blanks	300 ±60	0 (negative value)
SNI-21	W-2412	Upper midden	?	50cm (approx.)	Haliotis cracherodii	2070 ±200	1600 (1365) 1190
SNI-21	W-981	Lowest midden	?	160cm (approx.)	Haliotis rufescens	5070 ±250	5449 (5124) 4819
SNI-40	UCLA-147	Two burials	?	?	Olivella beads	3980 ±100	3819 (3676) 3549
SNI-51	UCLA-197	Cremation	?	35cm	Charcoal	2440 ±80	2719 (2472) 2351
SNI-51	UCLA-195	Midden	?	180cm	Mytilus californianus	2550 ±80	2039 (1929) 1833
SNI-51	UCR-713	?	?	190cm	Shell	2600 ±100	2119 (1992) 1870
SNI-51	UCR-714	?	?	198-213cm	Shell	3170 ±90	2786 (2727) 2670
SNI-51	UCLA-196	Midden	?	210-225cm	Haliotis cracherodii & Mytilus californianus	3170 ±90	2786 (2727) 2670
SNI-339	UCLA-2728A	?	?	?	Mytilus	8215 ±85	8505 (8403) 8350

The site is complex, with four mounds (A,B,C,D); three strata (I, II, III) are represented in Mounds B and C. The strata are separated by light-colored sand with few cultural materials. All three strata had surface expressions on sloping sides and eroded tops of the mounds.

All of the dates are from charcoal samples, which leaves us with at least three possible explanations: (1) some of the charcoal may have been contaminated with asphaltum which would produce an older date; (2) the charcoal samples were not in their original stratigraphic context; and (3) the stratigraphy of the site is not as undisturbed as it appears. Additional radiocarbon samples, obsidian sourcing, and soils and geomorphological studies are needed to resolve these issues.

Bleitz (1991) provides an analysis of a basketry impression found in a lump of asphaltum recovered from Stratum III, Mound B, at a depth of 90 cm. The impression is from a basket which was made by the coiling technique. The coiling was clockwise. This find may push the introduction of coiled basketry into California back earlier than 2000 B.C.

Salls (n.d.b) analyzed the fish remains from SNI-11 as part of his dissertation research. Two test units were analyzed, one from Mound A which contained materials assigned to Stratum I, and one from Mound B which represented Strata II and III.

The intensity of fishing at this site resembles that of modern commercial fisheries. A total of 43,404 specimens were recovered, the equivalent of 5,293 per cubic meter. The total minimum number of individuals was 2,193. The fish specimens per cubic meter are more than five times as many compared to what may be the densest site on San Clemente Island, Eel Point (SCI-43C), even though that sample was screened with 1/8-inch mesh compared to the 1/4-inch mesh used at SNI-11.

The exceptionally wide diversity of fishes recovered from SNI-11 (some 70 species) indicates a non-selective fishery. There does not appear to be a preferred species such as was documented for San Clemente Island where sheephead was the most common species at every site examined (Salls n.d.b). The fish sample from SNI-11 indicates a watercraft-oriented strategy where hook-and-line fishing was extremely important – many were rock and kelp fishes. Dip nets, drift or gill nets, and beach seines were also probably used, as indicated by the presence of small schooling fish such as sardines and silversides.

The analysis disclosed that 78% of the bones were from the head, while only 22% were other body parts. One possible explanation for this anomaly is that preparation and cooking methods may have exposed the vertebrae to differential preservation rates. Many large bowls, mortars, and pestles have been recovered from the island. It has been proposed that the grinding implements were used for processing fish. An alternate interpretation is that the fish were prepared for off-site or off-island consumption, in which case the heads were removed for transport to reduce bulk.

*Table 1. Calibrated ages were calculated using the CALIB program (Stuiver and Reimer 1986). Calibrated dates show the extremes of the one-sigma range with the calibrated ages between them in parentheses. For brevity, a range of calibrated ages is shown where three or more ages were actually calculated. All calibrated dates use the dating error as reported by the individual laboratories. Charcoal dates were calibrated using a bi-decadal data curve (Stuiver and Pearson 1986; Pearson and Stuiver 1986; or Pearson et al. 1986 as referenced in Stuiver and Reimer 1986). Marine samples were calibrated using the bi-decadal marine radiocarbon model (Stuiver, Pearson, and Braziunas 1986), using the Delta R of 225 ± 35 as recommended by Stuiver, Pearson, and Braziunas 1986 (as referenced in Stuiver and Reimer 1986).

Eighteen otoliths from SNI-11 were examined for seasonality data (Huddleston n.d.; Salls n.d.b). Seasonality readings were obtainable on all specimens. Seventeen of the otoliths, representing 94% of the sample, were captured in the winter season, from approximately early October to mid-May. Only a single specimen was from a fish caught in the summer. The 17 otoliths with rings indicating winter capture were subdivided into fall and late winter/early spring periods. Eleven (65%) were fall (early October to the end of December). The remaining six fish were caught during the late winter/early spring (early March to mid-May).

Analysis of the mammal and bird bone (Bleitz-Sanburg n.d.a) indicates an emphasis on the hunting of fur-bearing marine mammals. There is a gradual decrease over time in the acquisition of large pinnipeds, such as elephant seals and sea lions. Although the quantities do not indicate that they were a primary focus, there was a marked increase in the procurement of dolphins and porpoises.

The complex of birds identified at SNI-11 suggests a pattern of exploitation centered offshore and around the kelp beds, which was incidental to a more generalized fishing procurement strategy. The range of bird species appears to diminish in the Late Period.

A wide variety of mammals is characteristic for the Late Period. The distribution of mammalian fauna reflects an insular shore-based collecting and hunting pattern, which was dominated during the late occupation by an intensified exploitation of offshore fauna foraging in and around the kelp beds.

Lithic analysis was undertaken on 7,000 debitage flakes recovered from Mound B (Clevenger n.d.). Based on this analysis, it is proposed that the lithic manufacturing strategy represents an attempt by the inhabitants of the island to make use of locally-available lithic materials and limit dependence on off-island resources. Clevenger determined that they were using relatively small, coarse-textured metasedimentary and metavolcanic cobbles, available in conglomerate bed exposures on the island, as a source of raw material rather than imported lithic material of better quality. The island is formed of sedimentary rock, and the materials favored by flintknappers, such as cherts and volcanics, are not available. Clevenger describes the tool production method as a "split cobble reduction strategy" using a bipolar flaking technique. The tools made in this manner, while rather crude in appearance, were evidently adequate for the processing of fish and shellfish.

Obsidian hydration measurements were made on two small flakes from this site (Table 2). Interpretation of the measurements is problematical because the obsidian was not sourced and the samples were not recovered from the same or equivalent depositional contexts as the radiocarbon samples. If, based on information from the mainland (e.g., Orange County), we were to make the assumption that the obsidian is from Coso, there is a considerable range of age estimates for hydration rates proposed for Coso obsidian. For example, the range for five microns is from 712-3125 years B.P. (Hall 1988:37). Interpretation may be possible in the future through the dating of carbon, which may be available in curated column samples recovered from the same depositional contexts as the obsidian. Lacking this information it is not possible to say more than that the hydration rims of 5.4 and 7.7 are consistent with the stratigraphy.

Table 2. Obsidian Hydration Measurements from San Nicolas Island

Site	OHL No.	Catalog No.	Unit	Depth	Source	Hydration
SNI-11	7776	19-534	9.5S/3E	0-10 cm		5.4 μ
SNI-11	7777	19-515	1.5S/58.5W	40-50 cm		7.7 μ
SNI-16	7350	323-111	Pit 3, Mound C	0-6 in		3.3 μ
SNI-16	7351	323-114	Pit 3, Mound C	6-12 in	Coso	6.5 μ
SNI-16	7338	13-298	6S/18W	50-60 cm	Coso	6.8 μ
SNI-16	7339	13-334	1.5N/36E	60-70 cm	CasaDiablo	6.2/6.9 μ
SNI-16	7340	13-314	3.5N/27E	10-20 cm	Coso	7.1 μ

SNI-14

SNI-14 is located on the northwest coast of the island. It is a sparse shell midden, approximately 165 by 105 m in extent and 10-50 cm deep, capping a large sand dune. The site is situated near a rocky shoreline area. SNI-14 was test-excavated by the UCLA Archaeological Survey in 1959 to recover an exposed burial (Reinman and Townsend 1960).

The burial was a primary inhumation of an infant. It was tightly flexed and lying on its right side. The orientation was undetermined. No pathologies were noted. Metric data are available in Rootenberg (1960). Accompanying the burial were 11 spire-lopped olivella beads and a bone implement made from a mammal long bone.

SNI-15

SNI-15 is situated above a rock intertidal shore on the northwest coast of the island. It consists of patches of midden on two large dunes connected by a low saddle. The occupation debris is 10-20 cm deep and extends over an area 260 by 140 m. SNI-15 was test-excavated by the UCLA Archaeological Survey in 1959 to recover an exposed burial (Reinman and Townsend 1960).

A single burial was found weathering out of the sand at the base of the dune. It was a primary inhumation of a young adult female in a tightly flexed position, lying on the right side. The skull exhibited some degree of occipital protrusion or "bunning." Metric data are available in Rootenberg (1960). Associated with the burial were 568 tubular bird-bone beads in the chest and neck regions, perhaps indicating a necklace. Several of the bone beads were incised. A steatite bead and a bone awl were also with the burial.

SNI-16

SNI-16 is located on the northwest coast of the island. It is a typical dune site, consisting of patches of shell midden capping a large dune approximately 2,580 m in length and some 25 m high. Observable midden covers an area of about 428 by 123 m. SNI-16 has been tested on three different occasions (Reinman and Townsend 1960; Reinman 1964; Lauter n.d.b).

A burial area was excavated by the UCLA Archaeological Survey in 1958 (Reinman and Townsend 1960). The burial area was located on the landward end of the dune. It was approximately 15-20 m below the midden remains on the top of the dune and followed what would seem to be the preferred burial pattern in that it was off the main midden in a fairly level area.

Forty-three 5- by 5-foot squares were excavated to depths ranging from 24 to 36 inches. Much of the excavation area had been previously disturbed by relic hunters. The most

productive area was located in the most heavily disturbed portion of the site. In this area, in addition to Burials 2 and 3, a large number of abalone and mussel shell pendants of many types, stone and bone pendants, and large numbers of scattered human and animal bones were found. Scattered among the bones, pendants, and burials were many small basketry fragments containing red ochre, as well as a number of beads of steatite and shell. On the surface were scattered a number of skull and long-bone fragments. In addition to these, a number must have been destroyed by the extensive erosion that has taken place on this end of the island.

In all, three burials were recovered that could be considered undisturbed. Burial 1 was a young female buried face down, tightly flexed. The burial exhibited exostosis of the auditory meatus and occipital protrusion or "bunning," congenitally-missing mandibular third molars, and an abscessed tooth (Rootenberg 1960). No artifacts were recovered with this burial. Burial 2 was a male in a tightly-flexed sitting position with the skull partially protruding through its covering of asphaltum-coated matting or basketry. The hands were covering the face. Several steatite and bone beads and pendants accompanied this burial. Burial 3 was an infant either wrapped in asphaltum-coated matting or placed in a basket. It was unaccompanied by artifacts. Metric data for these burials are available in Rootenberg (1960).

A total of 394 artifacts was recovered from this excavation. Of this number 147 were shell pendants. Most were unassociated with the recovered burials but were found in the other excavation units. A number of them can be attributed to burials previously destroyed by erosion, but many were also found in various units either out of the disturbed area or below the level of disturbance. In several of the units human mandibles, skull fragments, and long bones were found in situations of partial articulation as well as in complete disorder. Some of these bone assemblages might possibly represent the practice of reburial. Excavation to a depth of 36 inches to sterile sand failed to reveal any deeper burials.

Additional work was performed at the site in 1962 (Reinman 1964) to explore the midden area. The site was divided into four separate mounds (designated from the seaward end to inland as Mounds A, B, C, and D). Midden remains are found on each of the four mounds and each was tested during the excavation. Eleven 5- by 5-foot pits were excavated (one in Mound A, four in Mounds B and C, and two in Mound D). The maximum midden depth of 36 inches was found on Mound A and the shallowest on Mound B, where the midden ended between 0 and 6 inches from the surface. In addition to the excavation of the units, column samples were taken from each mound to provide data on the midden constituents. Two radiocarbon samples were processed (Table 1).

The total number of artifacts recovered in the excavations, including those gathered from the surface, was 138; 98 were stone, 25 bone, and 14 shell. In addition, a single basketry impression was found on Mound B. A total of 19 cubic yards of midden was excavated, with the greatest amount taken from Mound C.

The most recent excavations at SNI-16 were directed toward establishing a chronology specific to the island (Lauter n.d.a, n.d.b). The excavations centered on known midden areas and artifacts noted on the surface. Eight 1.5- by 1.5-m units were excavated.

The midden consisted of several strata that were identifiable by their color. Abalone shells and shell lenses occurred frequently in all test pits. Mussel, tegula, land snail, and sea urchin were abundant in every pit and at most levels. Tegula, sea urchin, and land snails were found primarily in pockets or lenses and may represent a single processing event. These pockets are distributed throughout the site and occasionally may be seen eroding out of the

surface or sides of the dune in small piles. Chiton, limpet, norissia, olivella, and astrea were found in all pits as were fish, bird, and sea mammal bone. Artifacts total 708; 347 are utilitarian objects and 361 are ornamental.

Obsidian hydration measurements were made on three flakes found in the excavations and two from previous excavations (Table 2). Three have been sourced to Coso and one to Casa Diablo (Janet Scalise, personal communication, 1993). As mentioned previously, accurate interpretation of the measurements would require sourcing of the obsidian and the dating of carbon from the same depositional contexts as the obsidian.

One burial was recovered in the 1979 excavations. The burial was a primary inhumation, positioned on its left side, loosely flexed, and oriented toward the north. It has been identified as a probable female about 35-40 years of age with no evident pathologies (Salls n.d.a). Associated objects included a side-notched projectile point fragment, a sea mammal tooth pendant, a sea mammal effigy, a clam shell ring, a worked clam shell object, and three steatite pendants, two of which had biconical perforations. The remains were in very poor condition due to natural decomposition.

SNI-18

SNI-18 is located near the northwest tip of the island with major rocky intertidal zones both to the east and west sides of the site. These adjacent areas currently are a major rookery for western gulls, Brandt's comorants, and California sea lions. It is a very high dune site, about 700 by 175 m, consisting of shell midden scattered over the top and sides. It is probably the largest remaining site on the island. Excavations were undertaken by the UCLA Archaeological Survey in 1959 centering on an exposed burial area (Reinman and Townsend 1960).

Burial 1 was observed weathering out of the sand on top of the dune. Burials 2 and 3 were discovered about 100 feet east, also eroding out of the midden. Burials 2 and 3 had been disturbed. The skeletons were not articulated, and the bones ranged in depth from the surface to 35 inches. It is possible that they are reburials that were disturbed when the graves were dug for Burials 4 and 5, which were immediately below. The only artifact associated with Burial 4 was a large ball of yellow ochre in the pelvic region. Burial 5 was slightly below this burial and somewhat to the west; it was evidently the cause for the disturbance of the previous burials. It was an adult male, tightly flexed, lying on the left side, with several artifacts in association. Included were a hafted point, four complete abalone fishhooks, a stone pipe, and a possible "treasure box" consisting of four interlocking abalone shells that contained an incised steatite pebble. The "treasure box" was situated directly over this burial but at only 8 inches from the surface, while the skeleton was at a depth of 36 inches.

Burial 2 exhibits exostosis of the auditory meatus. Burial 4 exhibits occipital protrusion or "bunning." Burial 4 also has an osteoma at the juncture of the left parietal, frontal, and sphenoid sutures (possibly caused by an old injury), and on the rear of the left parietal along the left lambdoid suture. Burials 2, 4, and 5 had abscessed teeth. Metric data are available in Rootenberg (1960).

In excavating the burials, two hearths were uncovered. The first was 4 feet in diameter and 17 inches thick, extending from 17 to 34 inches from the surface. Within the pit at the 34-inch level was a series of stones in a triangular arrangement; within this area was the greatest concentration of ash, shell fragments, pieces of asphaltum, a large quantity of sea urchin spines, and a few scattered artifacts. The latter included a stone pestle, a bone awl,

pieces of rock with asphaltum, and scattered burnt animal bone (sea lion, bird, and fish bones as well as badly charred pieces of whale vertebra). The second hearth was a double fire pit, 91 inches in length, 58 inches wide, and in the form of two overlapping circles. It too contained ash, shell fragments, sea urchin spines in great quantity, pieces of asphaltum, and a large number of artifacts. Included were a stone pestle, two cores, a chopper, a bone wedge with asphaltum, a stone net sinker, two bone awls, a bone spatulate object, and a broken stone slab covered with asphaltum.

No further burials were recovered, and only a few artifacts were recovered from the other pits, including a bone gorge, the broken tip of an awl or flaker, and some fishhook fragments and blanks. A total of 128 artifacts were recovered from the excavations. One shell sample from the burial area was submitted for radiocarbon analysis (Table 1).

Two dog burials were also found, both completely articulated. They were not accompanied by artifacts.

SNI-38, Daytona Beach Site

The Daytona Beach site is located on the southeast coast of the island adjacent to a large sandy beach. There is a rocky intertidal area just to the east. The site was heavily impacted when the current barge landing was constructed. The remaining site area covers some 160 by 140 m. The site contains a wide variety of cultural materials.

Rozaire undertook excavation of SNI-38 primarily to learn more about weaving skills of the islanders (Rozaire 1959, 1967). Six 5- by 5-foot squares were excavated. Midden depth was a maximum of four feet. Two possible occupations are represented by differing colors of soil, the lower layer being decidedly more brown than black. The side-wall profile of one unit revealed a sterile overburden of three feet. The excavated portion of the site has since been destroyed to make room for the current barge landing, although some deposit still remains to the east. Woven sea-grass specimens, cordage, and limited evidence of basketry impressions were found.

Over three dozen examples of woven sea grass were recovered. All examples have plain, two-element S-twining. The warps and wefts were essentially small bundles of untwisted sea grass. About 20 examples of twisted cordage made from sea grass were also recovered. Structurally, most were S-strand with two Z-yarns. These findings were consistent with Rozaire's (1957) earlier proposal that twining techniques are culturally distinctive and that S-twining is associated with Uto-Aztec peoples.

A feature consisting of 130 tarred pebbles was recovered near the only burial recovered during the excavations. The feature measured 33 by 25 inches and was at a depth of 15 inches. A second concentration of 19 tarred pebbles was recovered at a depth of 26 inches in another pit.

The excavations yielded 75 stone artifacts, 23 bone artifacts, 61 fishhooks, 19 fishhook blanks, 10 shell beads, 4 specimens of woven sea-grass, and 7 of cordage. Numerous shellfish species, crustaceans, sea mammals, fish, and bird remains were also recovered.

One burial was recovered. It is possible that it was in a flexed position. It appeared to have been placed in a small depression with a considerable amount of charcoal. From the condition of the skeleton, it is assumed that the burning in the pit occurred prior to interment. Scattered within the pit were fragments of woven sea-grass and some small concentrations of unwoven sea-grass.

A second burial was later recovered at or near SNI-38 in 1960 (Rozaire 1970). The burial was exposed in a cut bank about 914 m from the shoreline. No midden soil was associated with the burial though scattered shell was observed on the surface nearby. The skeleton was contained within an oval grave pit which measured about 58 by 94 cm and was situated a maximum of 84 cm below the surface. The burial was tightly flexed, in a ventral position, and face down with the head pointing to the south. The individual was a male in his early twenties. The two hemispheres of the frontal bone had not united, thus producing a metopic suture from bregma to nasion. The only associated cultural materials were a few bits of woven sea grass and fragments of cordage which may, together, constitute the remnants of a cape or shroud.

SNI-40

SNI-40 is located near the west coast of the island overlooking a rocky intertidal area which is currently used heavily by sea lions. The site area measures some 300 by 100 m. Depth of the observed midden is in excess of one meter. A number of exposed burials were recovered by the UCLA Archaeological Survey in 1959 (Reinman and Townsend 1960).

The dune itself is about 805 m in length, running from northwest to southeast. It is about 61 m wide and perhaps 23 m high. The burial area was on the inland slope of the dune in a level space 3 to 6 m below the top of the southwest side. The burials were all found in sterile sand with the midden seemingly skirting the edge of the area and continuing on beyond for about 15 m, but only on the eastern side.

A 30-foot trench was laid out in a north-south direction over an area disturbed by previous looting. This was later extended into a larger grid encompassing some 88 five-foot squares, of which 57 were excavated, and from which 20 burials and 136 cubic yards of earth were removed. With the burials were found 75 artifacts, exclusive of 5,384 shell beads of varying types. Twenty-five of the artifacts were abalone shell dishes. A sample of olivella beads was submitted for radiocarbon analysis (Table 1).

Since the burial area was in sterile sand, no stratigraphy was noted, with the exception of three features. All three consisted of a burned layer of sand, charcoal, and large quantities of land-snail shells.

Burials 1 and 12 evidenced exostosis of the auditory meatus. Burial 15 exhibited several wormian bones along the lambdoid and possibly the sagittal sutures. Burials 4 and 9 exhibit occipital protrusion or "bunning." Burials 1, 3, and 4 have congenitally missing mandibular third molars. Burials 4 and 16 have dental abscesses. Burial 6 was missing four teeth, possibly due to an accident. Several teeth were missing from Burial 9 indicating periodontal involvement. Metric data are available in Rootenberg (1960).

SNI-51, Dutch Harbor

The Dutch Harbor site is located at the west end of Dutch Harbor, near the center of the southern shore of the island. The site covers an area of about 325 by 100 meters. Dutch Harbor is the largest sandy beach on the island. The site is located near the western end of the beach, and there is also a rocky intertidal area nearby. The site was tested in 1958 (Rozaire 1959, 1967), with the primary intent of recovering woven material, and in 1960 to obtain datable materials in stratigraphic context (Rozaire 1970).

Fourteen pieces of woven sea grass were recovered and all but two have some remains of selvage. The basic weave is plain two-element S-twining, with braided three- and four-element S-twining found in the top selvage of four specimens. Over 20 pieces of twisted sea grass cordage were uncovered. All were S-strand, with two Z-yarns, except for one piece of S-strand with two S-yarns. These findings were consistent with Rozaire's (1957) earlier proposal.

The original excavation consisted of eight 5-foot squares. The squares were excavated to a depth of only 36 inches due to limited time. A bulldozer cut revealed that the midden extended to a maximum depth of nine feet below the surface.

Twenty stone artifacts, three bone artifacts, and 10 fishhooks were recovered along with 16 specimens of woven sea-grass. Numerous shellfish species, sea mammal, fish, and bird remains were also recovered.

The 1960 excavation consisted of four 5-foot squares. Two were excavated to sterile soil and two were terminated at depths of 30 and 54 inches due to time constraints. A portion of a cut bank was excavated to obtain some woven sea grass eroding from the cut. The woven sea-grass was encountered at a depth of about 6 feet in association with sea-grass cordage, a bone tool fragment, two scrapers, and charcoal.

The four excavated units revealed a clear stratigraphic profile which had numerous lenses. The profile probably reflects the slow build-up of the midden in an active dune area. Numerous artifacts were recovered from the surface as well as from below the surface. In general, artifact yield greatly decreased below 48 inches.

Two burials were also recovered at this time: one inhumation and one cremation. The cremation was found eroding out of the side of a cut bank 14 inches below the surface and several hundred yards above the road. The fragmented and charred bones were contained in a saucer-shaped depression, 35 by 24 inches, which had been excavated seven inches into the compacted yellowish sand. The soil in the bottom of the pit was slightly reddish and was covered by a thin layer (less than 1/2 inch thick) of chunks of charcoal. Charcoal and dark earth filled the rest of the pit; it was in this matrix above the thin charcoal layer that most of the fragmented human bone occurred. Though the bones were carefully exposed, there was no apparent pattern to give a clue to original position. It would appear that the remains constitute a secondary burial following cremation. In the course of burning, the bones had been poked around to assure that they were completely charred by the fire. Over 1,300 shell and bone beads were scattered throughout the pit. These appear to have been thrown loosely into the cremation fire during burning rather than as a necklace. Interestingly, over half of the beads were not charred. The cremation pit also contained a few small chunks of red and yellow ochre and five pieces of small rock, shell, and animal bone fragments. Approximately 100 tarred pebbles were scattered about in small clusters of three or four above the cremation. The cremation was associated with the remnant of a shallow, compacted, dark-colored midden 10 to 14 inches thick (Rozaire 1967, 1970).

The inhumation was discovered about 25 feet north of the cremation, weathering out of a small gully about two feet deep. The bones were disarticulated, broken in some instances, and mixed and scattered over an area of about 41 inches square, indicating a secondary burial. The skull was upright facing west. Some small scraps of woven sea grass and fragments of cordage were associated with the bones (Rozaire 1970).

Charcoal from the cremation and shell samples from an unknown location within the excavated areas were submitted for radiocarbon analysis. The radiocarbon determinations appear to be stratigraphically consistent (Table 1).

SNI-56, Redeye Site

SNI-56 is located on the northwest coast of the island in an area known locally as Redeye. It is adjacent to a small sandy beach with rocky intertidal areas to the east and west. A number of water sources are nearby. The site extends over a 300 by 175 m area of grassy hummocks and patches of midden. It displays an extremely wide array of cultural materials. Three burials were excavated by the UCLA Archaeological Survey, one in 1952 and two in 1959. A small surface collection was also made by Reinman and Townsend (1960).

Burial 1, with the exception of the skull, was not salvaged. The skull was resting on a spool-shaped object made from a whale vertebra (Meighan and Eberhart 1953). The 1959 excavation centered on two exposed burials and consisted of eight 6- by 6-foot units. Burials 2 and 3 were found weathering out of the sand in an area down from the main dune area in the same location from which Burial 1 was recovered in 1952. Other artifacts found with the burials at this site include the following: Burial 2 had five unmodified abalone shells, and Burial 3 was accompanied by a few spire-topped olivella beads, a perforated stone ring, four unmodified abalone shells, and three fish gorges of bird and mammal bone.

Burial 2 was a young adult female. This was a primary inhumation in a tightly flexed position lying on the left side and oriented toward the west. A cylindrical growth was observed extending downward between the right mastoid and the right occipital condyle. The growth has an articular facet on the anterior portion and may have articulated with the transverse process of the atlas. If so, it would have limited vertical head movement and horizontal movement to the left. The skull exhibits occipital protrusion or "bunning." The dentition exhibits congenitally-missing third molars (Rootenberg 1960).

Burial 3 was the primary inhumation of a young adult male. It was in a tightly flexed supine position oriented toward the north. The skull exhibits occipital protrusion or "bunning." Metric data for both burials are available in Rootenberg (1960).

SNI-79, Powerhouse Site

The Powerhouse site is located on the central plateau at about the center of the island. Celery Creek, one of the major drainages on the island, is located just to the west. The site was tested by the UCLA Archaeological Survey (Reinman 1964).

SNI-79 is a rather large site, covering a slightly raised knoll some 370 m long and 115 m wide. Crude core tools of various kinds as well as bowl fragments, pestles, and other implements were found scattered over the surface of the midden. The midden consists of a lightly compacted sandy soil overlying a lighter-colored sandy layer and is only slightly darker than the surrounding soil. Small shrubs and grasses cover the surface, making a survey of the site difficult.

Five 5- by 5-foot test pits were excavated to a depth of 24 inches. Three of the pits (1, 3, and 5) were screened through 1/8-inch mesh, with all of the material from Pit 3 saved for laboratory analysis (Reinman 1964).

A total of 143 artifacts was recovered, including the surface collection. Nine cubic yards of midden were excavated. This gives an average of 14.5 artifacts per cubic yard (excluding

the 12 from the surface collection). The catalog from this site reveals a relatively high percentage of stone artifacts compared to the other sites excavated. The amount of shell in the midden was also considerably less than at other sites, with only 80.3 grams of shell per cubic foot of midden.

Both fishhooks and hook blanks were recovered from all levels of the midden. In addition, a few stone tools (drill, reamer, file) which were apparently used in the manufacture of fishhooks were also found. There were also a number of pestles and bowl fragments on the surface and in the excavations. Two small fragments of bone were clearly tools, either a wedge or pry fragment and a broken pointed object, probably an awl or flaker. Two other bone fragments, one covered with asphaltum and the other a small, incised fragment, constitute the remaining bone artifacts. Of the 26 shell objects, 22 are either fishhooks or fishhook blanks. The remaining four artifacts include a perforated norissia shell and three fragments of abalone covered with asphaltum.

Radiocarbon samples have not been submitted from SNI-79; however, one fishhook has been identified as a Late Period type (Strudwick n.d.) which may indicate that the site was occupied during this period. Five major food shells were identified: abalone, mussel, tegula, sea urchin, and limpet.

SNI-351

In 1991 and 1992 the California State University, Los Angeles archaeological field school conducted test excavations at SNI-351, a shell midden and lithic site situated on the central plateau. This site was selected to provide information that can be compared with the shoreline sites, SNI-11 and -16.

SNI-351 has been affected by both wind and water erosion. Portions of the site are dissected by ravines, and there are large areas that have been deflated and resemble a desert pavement of lithics, pebbles, and caliche.

The main cultural deposit consists of a stabilized sand dune containing a shell midden. The dune midden deposit forms an L-shaped mound above a large ravine at the south-central portion of the site. The mound covers an area approximately 10.5 by 4 m in a generally east-west direction and 7.5 by 4 m north-south.

Additional midden deposits are found below the mound in sandy/silty soil to the east and north. These deposits do not extend more than 10 to 15 m beyond the central dune.

Another smaller shell midden sand dune is situated within the northwestern portion of the site. The depths of the midden deposits range from 50 to 110 cm.

A number of small, discrete shell deposits with little or no depth were recorded within the terrace extending toward the east. A series of shovel test pits extending from the midden area to the east were excavated and screened. These contained only surface material and helped to define the extent of the main midden area.

A total of 23 1.5- by 1.5-m units were excavated. The material was screened through 1/8-inch mesh. The residues left in the screen were bagged and sorted in the lab. This resulted in the retrieval of otoliths and small fish bones, beads, and flaked stone.

Preliminary analysis indicates that the bulk of the midden constituents consists of marine shell. The main shellfish types are abalone, mussel, tegula, and owl limpet. The majority of the bones are fish, followed by sea mammal. Morning glory seeds are plentiful and some are

burned. Artifacts include small amounts of asphaltum, including plugs; olivella beads and bead detritus; flaked stone cobbles, including bifaces; a few flakes and shatter of exotic materials such as obsidian, Monterey chert, and quartz; and a few bone tools. Although a few whole pestles, bowl fragments, a metate, and a mano were recovered from the surface, groundstone artifacts were rare within the subsurface deposit. Features consisted mainly of deposits of burned shell and bone, soil stains, and hearths.

SUMMARY

Some summary figures will help to put the current state of knowledge into some perspective. Five sites have been excavated for the recovery of exposed burials. Only five sites have been tested in terms of defining midden constituents.

SNI-11 is the best documented of the tested sites. However, there are still some areas of additional analysis that would prove fruitful. Additional radiocarbon samples should be processed and sourcing of the obsidian samples should be performed to correct inconsistencies and clarify the stratigraphic sequences. Geomorphological studies may provide useful data regarding the post-depositional processes of a sand dune midden deposit. Additional analysis of the reptilian, crustacean, and botanical remains should be conducted. A synthesis report should be prepared and published.

SNI-16 has been adequately sampled, though reporting of this site is incomplete. A detailed analysis of the faunal remains compared against the SNI-11 sample would be very instructive in defining shoreline activities and subsistence strategy. An attempt should be made to obtain radiocarbon dates from the same depositional contexts as the obsidian. A synthesis report should be prepared and published.

Additional sampling of SNI-79 would be very fruitful. Until recently, this was the only site on the plateau that had been tested. Additional sampling and analysis would provide information for comparisons with shoreline sites and with SNI-351, a recently excavated plateau site that appears to have more shell.

The analyses of the materials recovered from SNI-351 have not been completed. The data from these studies can be used to determine the type of site, length, season, and time of occupation, and to compare SNI-351 with the shoreline sites. The results of these analyses may provide clues to why this location on the central plateau was selected for occupation. The nearest water source may have been Celery Creek. This creek and its tributaries are dry today, but it is obvious that water played an important role in the formation of the numerous ravines that dissect the plateau. Shellfish processing and consumption appear to be the main activities conducted at the site, but since the site is not situated at a convenient distance from the ocean, it is probable that the shellfish were used to sustain the population while they were concentrating on some other activity. Some propositions to be evaluated are that the site was situated near an important plant resource or near a cobble conglomerate bed exposure. Other factors that may have played a part in the selection of this location are the elevation and view of the other islands and the practical consideration that the site would be free of kelp flies and other coastal pests that plague people while processing shellfish at the shoreline during certain seasons.

Environmental reconstruction is needed using geology, geomorphology, paleoclimate, soils, flora, and fauna analysis. Sourcing the origins of the artifacts and the flora and fauna remains, and determining the means and the energy expenditure required to get them, is another important line of inquiry.

Additional testing of SNI-38 and SNI-51 should be conducted to provide an adequate sample. This, coupled with additional research on existing collections, would provide solid comparative data to examine differences between the northwest shoreline, plateau, and south shore.

The other sites have only been tested in terms of burial recovery and thus do not provide any midden materials that could address archaeological research questions such as cultural change, diet, ecological adaptation, and environment. In addition, a number of other sites on the island have been excavated but not reported. These include investigations by Malcolm Rogers and George Carter in 1930, Arthur Woodward in 1939-1941, and Phil Orr in 1941. The collections and notes from some of the unreported investigations are still in existence, and a more complete analysis of these museum collections would add substantially to our understanding of San Nicolas Island prehistory. A synopsis of Malcolm Rogers's work from the 1930s will serve as an indication of this untapped potential.

Under the auspices of the San Diego Museum of Man, Malcolm Rogers, accompanied by George Carter, spent one month on the island in 1930 surveying and collecting. The team collected 45 cases of specimens (including 33 burials) which were taken to the San Diego museum where the collection was cataloged. Only a preliminary manuscript describing the fieldwork was completed (Rogers n.d.), though his field notes are on file at the Smithsonian Institution in Washington.

Table 3. Summary Data

Site units (5- by 5-ft or equivalent)	SNI-11	-14	-15	-16	-18	-38	-40	-51	-56	-79
Midden area	15	0	0	19	0	6	0	10	0	5
Burial area	0	1	1	43	13	0	57	0	11	0
Artifacts	0	12	570	1,102	128	?	75	?	>14	143
Burials	0	1	1	4	5	2	20	2	3	0

Chronology

Lauter (n.d.a, n.d.b) describes the island's chronology in terms of three cultural phases (Early, Intermediate, Late). At that time (1982) there were no dated sites that could be designated as Early Phase, pre-3000 B.C. The Early Phase includes the Paleo-Indian Tradition, ca. 13,000 to 11,000 B.C., and the subsequent culture complexes in southern California known as the San Dieguito Tradition, the Hunting Period, the Encinitas Tradition, the Millingstone Horizon, and the Early Period (Warren 1968, Wallace 1978, C. King 1981). This was the time of adaptation to the environmental changes of the Holocene, followed by regional developments emphasizing use of plant foods and the first use of marine resources (Moratto 1984). The earliest sites on the island are dated to the Early Phase. These include SNI-11, Mound B, Stratum III; SNI-21; and SNI-339.

The Intermediate Phase, ca. 3000 to 1000 B.C., includes SNI-40, SNI-16, and Strata II and III of SNI-11. This phase corresponds to Warren's Encinitas Tradition (1968), Wallace's Period III: Diversified Subsistence (1978), and King's Middle Period (1981). It is characterized by the introduction of the mortar and pestle used in the processing of acorns

and other hard seeds. The atlatl is used for hunting, and marine resources are exploited along the coast.

Late Phase sites, ca. 1000 B.C. to historic contact, include SNI-51, SNI-18, and Stratum I of SNI-11. Late Phase sites can be related to Warren's Shoshonean Tradition (1968), Wallace's Late Prehistoric Horizon (1955), and King's Late Period (1981). Late Period populations are characterized by a diversified subsistence, increased population density, reliance on maritime resources, social complexity, trade, and large permanent communities. The introduction of the bow and arrow, pottery, a monetary system, and artistic elaboration also mark this period (King 1981, Wallace 1955, Warren 1968).

In 1988, a number of additional radiocarbon dates were processed from SNI-11 (Breschini et al. 1992). These dates introduce an element of confusion regarding the chronology and stratigraphy of the site (Table 1). One date from the 140-150 cm level of Stratum III is 5033-4729 B.C., which falls within the Early Phase. This suggests an earlier occupation of the site than previously estimated. However, consideration must be given to the fact that this date is problematic because it overlies Stratum III, 170-190 cm, which has a later date of 2919-2505 B.C.

FUTURE DIRECTIONS

Islands are a natural laboratory for the study of cultural and biological systems and human adaptation. The ocean provides a barrier to many forms of terrestrial life, the physical boundaries are clearly delineated, and the island provides specific circumscribed environmental constraints and opportunities. The Channel Islands have abundant resources that were important to human populations. They are discrete geographic units with significant variation in the nature of resources and provide an ideal context to test hypotheses about cultural-environmental interactions.

San Nicolas Island provides a particularly favorable environment for archaeological research. As one of the smallest of the Channel Islands and the farthest from the mainland, it is a discrete geographical unit with sufficient resource and environmental variation to test hypotheses about human adaptive systems. For example, it is likely that the entire settlement system is represented in the archaeological deposits, and because the island is small, it may be possible to identify all of the components and to examine change through time. Although there has been some damage due to erosion, construction, and collecting, archaeological preservation is generally good. An important plus is that because of the absence of burrowing animals, stratigraphic contexts are largely undisturbed.

San Nicolas Island was remote (there is no evidence of a land bridge), small, wind-swept, semi-arid, sparsely vegetated, and the only terrestrial animals are the island fox, a mouse, and a lizard. Assuming that these conditions are approximately similar to those of the past few thousand years, the island population clearly faced some environmental challenges. The materials recovered from the archaeological excavations can be analyzed to reconstruct the prehistoric environment and to address questions regarding prehistoric human adaptation.

The major research domains that have been identified for San Nicolas Island prehistory are as follows:

Paleoenvironmental influences

Recent paleoenvironmental research has produced a wealth of information regarding environmental change during the Holocene (Anderson n.d.; Schimmelmänn n.d.; Stine n.d.;

Wigand n.d.). Paleoenvironmental reconstructions can be used to identify the environmental factors that may have influenced hunter-gatherer adaptations on the island. This information can be used to address questions regarding environment and cultural behavior. For example, two models have been proposed regarding subsistence change on the Channel Islands. The environmental model attributes subsistence change to environmental factors such as sea temperature (Glassow n.d.a, n.d.b; Glassow, Wilcoxon, and Erlandson 1988; Walker and Snethkamp n.d.). The alternative model proposes that prehistoric exploitation strategies and the pressure of human predation on limited island resources provides the best explanation for diachronic changes in settlement and subsistence patterns (Bleitz-Sanburg n.d.a, n.d.b; Bleitz-Sanburg and Salls n.d.; Raab n.d.; Raab and Yatsko 1990; Reinman 1964; Salls n.d.b).

Settlement systems

We do not know the size of the prehistoric population on San Nicolas Island or whether it was occupied on a continuous basis. Unlike mainland territories where the boundaries are ill-defined and many parts of the settlement pattern have been lost, San Nicolas Island provides a relatively finite and manageable sample universe for the analysis of settlement patterns and population size. If it is assumed that the location of sites in a particular area reflects a pattern of adaptive behavior, the reconstruction of the settlement pattern can be used to examine a broad range of cultural and environmental factors that may have influenced hunter-gatherer adaptations on the island.

This line of inquiry is important. Although the locations of all of the sites that are visible on the surface have been mapped, additional investigations are needed to identify site function and to reconstruct the configuration of the settlement pattern. There is some information regarding material culture and faunal remains from a few of the coastal sites, but there is a lack of information about site function and how the aboriginal population used various areas of the island, especially the upper central plateau.

Subsistence strategies

There is a preponderance of data indicating a marine subsistence focus; however, very little is known regarding the subsistence patterns on San Nicolas Island. The emphasis has been on the investigation of coastal sites; little is known about subsistence on the central plateau, and even less is known about aboriginal utilization or availability of plant foods. For example, it is not known whether the large number of mortars and pestles found on the island were used for the preparation of plant materials or pounding fish and shellfish. (See also the discussion of paleoenvironmental influences.)

Island cultural chronology

Culture sequences have been formulated for southern California prehistory (Wallace 1978; Warren 1968; King 1981; Chartkoff and Chartkoff 1984; Moratto 1984). These can be used as a source of information to test hypotheses regarding the prehistory of San Nicolas Island.

Raab and Yatsko (1990) suggest that these reconstructions may not accurately reflect cultural development on the Channel Islands, particularly the earliest cultural periods. For example, radiocarbon dates and recent information regarding aboriginal fisheries on the Channel Islands suggest that aboriginal occupation began as early as 8,500 B.C. on San Miguel Island, 7,700 B.C. on San Clemente Island, and 6,200 B.C. on San Nicolas Island (Raab and Yatsko 1990; Salls n.d.a). Raab and Yatsko propose that these data reveal some of

the earliest fully maritime cultural adaptations in the Western Hemisphere and that maritime adaptations were an important aspect of coastal Paleo-Indian adaptation (1990).

According to Erlandson (1985), sea-level rise during the Holocene may have obliterated most of the evidence of early maritime adaptations along the mainland coast and encouraged a return to more terrestrial resources exploitation as the estuaries were drowned or filled with silt. Recent modelling by Pierson, Shiller, and Slater (n.d.) indicates that at 6550 B.C. San Nicolas Island may have had 25% more land mass than at present, and at 9050 B.C. the land mass may have been over three times the current size.

Based on a survey and collection of artifacts on San Nicolas Island in 1930, Malcolm Rogers proposed that three cultural periods are represented on the island. The early sites appeared to contain small quantities of marine shell and to lack bone or artifacts. As land snail shells were plentiful, he suggested that the earliest culture exploited them. Later the sites contain pink abalone, and house floors were observed between the piles of shell (n.d.).

The Canaliño period is the second cultural sequence described by Rogers. This period is characterized by many house pits of a size that would accommodate a family. Dogs were introduced to the island at this time. The first shell fishhooks appear, and midden constituents include black abalone, mussel shells, and a high percentage of fish bone. Later during this period, community houses measuring 30 to 40 feet in diameter appear. Six of these were described (Rogers n.d.).

The third period in Rogers's cultural sequence is the Shoshonean. This period is characterized by the presence of cremations, metates, obsidian projectile points, and perhaps the sweat house. Cremations were found in eight village sites near the west end directly over Canaliño-period deposits. Rogers suggests that the Shoshonean occupation does not date to any antiquity.

The culture sequences that have been proposed for southern California and San Nicolas Island can be tested using carefully collected, calibrated, and interpreted radiocarbon samples. The anomalous dates that have been recovered both from San Nicolas Island and San Clemente Island indicate that there are a number of problems to be solved. Since marine shell is more abundant than charcoal, it will be very important to solve the problems that interfere with the accurate dating of this material, such as contamination, upwelling, and the fact that different species vary in their habitat and ingestion of organisms in equilibrium with older seawater. Contamination of charcoal samples, perhaps with asphaltum, and the interpretation of radiocarbon dates in light of the complex stratigraphy of shell mounds also need to be addressed.

Social organization

Archaeological, ethnographic, and historic data indicate that at the time of European contact the Chumash and possibly the Gabrielino were organized at the chiefdom level of social complexity (C. King 1981; L. King 1982; Martz 1984). A general trend toward greater complexity is proposed for the Pacific Coast groups beginning around 2000 B.C. (Chertkoff and Chertkoff 1984). This social complexity included a system of social differentiation with hereditary political leaders at the top of the status ladder, followed by religious practitioners and craft specialists. This elite group controlled the procurement and distribution of surplus resources and had differential access to wealth. The rise of complexity is attributed to the high productivity of the Pacific Coast resource base as well as the social institutions that managed them.

It will be particularly interesting to address the question of whether a trend toward greater social complexity was present on San Nicolas Island, and to examine the basis for social organization, the level of complexity, and the development of this society through time.

Island-specific craft specialization industries have been identified on Santa Catalina and Santa Cruz islands (Arnold 1987, 1991; Rosen 1980). Because islands are relatively isolated and limited in resource variety, it is proposed that craft specialization and the development of a specialized manufacturing industry would be necessary to ensure adequate supplies of resources that are not available on the island. This would be a significant adaptive device and, as Arnold (1987) suggests, an impetus for the development and maintenance of social complexity.

Regional interaction and trade

Although ethnographic sources and historic accounts provide numerous descriptions of trade between the Channel Islands and the mainland, there is no information regarding the part that was played by the inhabitants of San Nicolas Island. Trade is very important to our understanding of San Nicolas prehistory because it involves the exchange of ideas as well as economic goods and can provide information regarding craft specialization and social complexity. In addition to artifact manufacturing areas indicating craft specialization and the production of surplus for trade, the presence of exotic plant and animal remains, raw materials, and finished artifacts would indicate trade.

Cultural affiliation

There is no mention of the Indians of San Nicolas Island in the diaries of the European explorers, and virtually all of the current ethnographic information regarding these people is based on the artifacts of the "Lone Woman of San Nicolas Island" at the time of her discovery. According to Kroeber, who analyzed four words spoken by the "Lone Woman," San Nicolas Island was occupied by Shoshonean speakers at the time of European contact (Eisen 1905). Based on archaeological and linguistic evidence, people speaking this language are thought to represent an intrusive desert culture group who displaced the Hokan speakers who were the original inhabitants of the Los Angeles, Orange County, Inland Empire, and southern Channel Islands region. The direction, nature, extent, and time of this intrusion is subject to debate with some placing the intrusion at about A.D. 750 and others arguing for an earlier occupation at about 1000 B.C.

Some possibilities regarding cultural affiliation of the populations occupying the southern Channel Islands are based on studies of human osteology on San Clemente Island (Titus n.d.). Titus suggests that two distinct genetic populations occupied the island. The earliest population is attributed to the widely distributed Hokan-speaking peoples and the more recent genetic population to the relatively recent Shoshonean speakers and ancestors of the Gabrielino people of the Los Angeles Basin and Santa Ana River watershed. The correlation between a genetic population and a linguistic grouping must be viewed with caution. However, it would be important to determine whether distinct genetic populations can be identified for San Nicolas Island and to reconstruct the sequence of occupation.

In addition, further analysis of preserved woven sea-grass and basketry impressions may reveal weaving techniques that can be attributed to a particular cultural affiliation (Rozaire 1959, 1967, 1970, n.d.). Variations in fishhook types may also provide important clues (Strudwick n.d.). If found in good stratigraphic contexts, this information can be compared with data regarding changes in settlement-subsistence strategies through time.

Technology

Very little is known about the technology of San Nicolas Island. Ethnographic information is not available, and the material culture must be reconstructed from archaeological data. According to early investigators, the sites contain an abundance of mortars and pestles, projectile points are rare, and metates are absent. Steatite effigies, shell beads and fishhooks, bone tools, and woven sea grass are also reported (Benson n.d.; Bryan 1927; Meighan and Eberhart 1953; Schumacher n.d.). De Cessac's observations in 1878 are particularly interesting. He noted that many of the artifacts are different from those he had encountered on the coast of Santa Barbara and the islands of Santa Cruz, Anacapa, and San Miguel. De Cessac was particularly impressed by the workmanship, originality, and beauty of the bird and fish effigies, the "stone vases," and the mortars and pestles. In addition to finished pieces, he evidently observed and collected items representing entire sequences of manufacture (Heizer 1951).

The flaked cobble stone-tool industry identified by Clevenger (n.d.) provides an interesting contrast to the groundstone craftsmanship described by de Cessac. Clevenger proposed that the inhabitants of SNI-11 were using locally-available cobbles as a source of raw material rather than imported lithic materials of better quality. Additional studies are needed to determine whether this lithic-procurement and manufacturing system is an island-wide adaptive system and represents an attempt to limit dependence on off-island resources, or whether it was confined to coastal fish- and shellfish-processing sites as the least expensive tool source. Further studies regarding lithic procurement and manufacture, stone tool use, and tool kits should provide additional insights regarding stone tool traditions and craft specialization.

The numerous fishhooks, fishhook blanks, fish gorges, cordage, drills, net gages, weights, and other related items recovered and observed on the island provide the opportunity to trace the development of fishing technology through time. The discovery of shaped and utilized shell fragments at SNI-38 opens another line of inquiry regarding the use of shell, perhaps as a substitute for wood (Dana Bleitz-Sanburg and Jane Rosenthal, personal communication, 1993). Shell bead and detritus features have been observed on the surface of various sites along the coast, and bead making may have been an important island industry.

Site formation and post-depositional processes

The formation of the archaeological sites on San Nicolas Island can be attributed to a variety of natural and cultural processes. In addition, natural and cultural post-depositional processes such as wind and water erosion, sedimentation, sand-dune formation, sheep ranching, and military activities have modified the sites and obscured the archaeological record. The patterns produced by these processes are often mistaken for cultural patterning; therefore a comprehensive understanding of their effects is critical for interpretations of site chronology, stratigraphy, spatial organization, and function (Erlandson and Rockwell 1987).

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